AI-Assisted Systematic Reviews and Meta-Analysis

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Abstract

The exponential growth of scholarly literature has made traditional, manual methods of evidence synthesis increasingly unsustainable across a wide range of academic disciplines. Systematic reviews, scoping reviews, and meta-analyses—once considered the gold standard for mapping research fields and evaluating interventions—now face significant bottlenecks in search, screening, and data extraction. Artificial intelligence (AI), particularly active learning and large language models (LLMs), offers powerful tools to accelerate these processes while maintaining methodological rigor. However, the scientific value of AI-assisted synthesis depends less on the mere act of automation than on transparent query formulation, rigorous bias mitigation, structured data extraction, and adherence to established reporting guidelines such as PRISMA 2020 and ROSES. This paper situates AI tools within a broader research workflow for evidence synthesis. It reviews relevant methodological literature on machine learning for abstract screening, LLM-based data extraction, programmatic literature collection via open APIs, and research ethics; discusses the opportunities and limitations of AI for literature reviews across disciplines; examines key problems of hallucination, structural bias, reproducibility limits, and human-in-the-loop oversight; and proposes a pedagogically oriented framework for introducing researchers to AI-assisted synthesis. Restricting the computational discussion to a single penultimate section, the paper outlines a workflow in which Python and R scripts hosted in a public repository support an interface-based application through which participants can query open APIs, train active learning models, configure LLM extraction prompts, and generate preliminary syntheses through a guided, no-code interface. The argument is that AI is best understood and taught as a powerful assistant within a rigorous methodological framework, requiring explicit validation and human verification at each stage.

Contents

1 Introduction

Systematic reviews remain one of the most time-intensive forms of research. Across the health and life sciences, the social sciences, the humanities, and engineering, researchers are expected to map rapidly evolving fields, evaluate interventions, and synthesize disparate findings into coherent, actionable conclusions. Systematic reviews and meta-analyses have long provided the methodological rigor required for these tasks, establishing themselves as the highest level of evidence in many disciplinary hierarchies. Yet, the traditional workflow—involving manual database searches, the screening of thousands of titles and abstracts by two independent reviewers, and the painstaking extraction of data from full-text PDFs—is often characterized by massive time investments, significant human error, and a growing mismatch between the pace of publication and the pace of synthesis.

The methodological consequence of this mismatch is clear: if evidence synthesis is to remain feasible and rigorous in an era of information abundance, the integration of computational and AI-assisted tools must be treated as a core component of research design rather than an optional convenience. The scientific community has begun to recognize this imperative. Funding bodies, journals, and methodological organizations are increasingly acknowledging the need for AI-assisted approaches, while simultaneously calling for transparency, validation, and the preservation of human oversight in the review process [Flemyng et al., 2025].

The emergence of machine learning and natural language processing has created unprecedented opportunities to automate the most labor-intensive stages of systematic reviewing. Active learning frameworks, such as ASReview, have demonstrated that human-in-the-loop algorithms can drastically reduce the number of abstracts that must be read to identify relevant studies, often achieving near-complete recall while requiring reviewers to screen only a fraction of the corpus [Van de Schoot et al., 2021]. More recently, the advent of large language models (LLMs) like GPT-4 and Claude has opened the door to zero-shot abstract screening and automated data extraction from full texts, tasks that previously required extensive human labor [Mahmoudi et al., 2025, Kim et al., 2025]. However, the central challenge is not simply the availability of these AI tools, but the capacity to align them with meaningful research questions, ethically acceptable screening strategies, and defensible validation procedures. Crucially, any such workflow must proceed with explicit validation and human verification at each stage.

Furthermore, the data collection phase of evidence synthesis is undergoing a parallel transformation. The rise of open bibliographic metadata catalogs, such as OpenAlex and Crossref, alongside programmable APIs, allows researchers to bypass proprietary, siloed databases and construct comprehensive corpora programmatically and reproducibly [Delgado-Quirós and Ortega, 2024, Culbert et al., 2025]. This broader framing is especially relevant because an AI-assisted review is only as good as the corpus it operates on. The researcher does not encounter a neutral, perfect set of literature, but rather a dataset shaped by indexing practices, API limitations, and the precision of search query formulation. Recognizing this dependency is the first step toward rigorous AI-assisted synthesis.

In adjacent work on text analysis, Grimmer, Roberts, and Stewart emphasize that data construction is inseparable from theory, representation, and validation; there is no value-free corpus construction, and the suitability of a method depends on the task at hand [Grimmer et al., 2022]. Those insights are highly applicable to AI-assisted systematic reviews. The movement from manual to AI-assisted synthesis is not a simple upgrade; it is a methodological shift that introduces new forms of efficiency alongside new forms of risk. The researcher who delegates screening to an active learning model or extraction to an LLM must understand the assumptions, failure modes, and validation requirements of those tools as clearly as a researcher who designs a survey instrument or chooses a statistical model.

This paper develops that argument in the context of AI-assisted systematic reviews and research synthesis across a wide range of academic disciplines. The aim is not to provide a comprehensive technical manual on specific LLM architectures or API wrappers. Nor is it to defend an unrestricted, fully automated approach to literature reviews that removes human judgment from the process. Instead, the goal is to provide a methodologically grounded account of how AI tools can be understood as distinct stages within a broader research workflow. That workflow begins with a substantive question and API-based programmatic literature collection, proceeds through active learning or LLM-based screening, and continues through automated data extraction, validation, preliminary synthesis, and ethical reporting. Understood in these terms, AI-assisted synthesis is less a magic bullet than a practical and epistemic bridge between overwhelming literature environments and rigorous, synthesized evidence.

The paper also has a pedagogical purpose. In many academic settings, especially in professional training for early-career scholars and advanced doctoral students across a wide range of disciplines, the demand for AI-assisted methods exceeds the coding experience of participants. This gap has important implications for how these tools are best introduced. If a seminar promises to cover the full technical universe of machine learning and LLM engineering in a short time, it is likely to be unrealistic and counterproductive. If, however, instruction is designed around a carefully delimited no-code workflow built on open scripts and an interface-based application, then participants can learn the logic of AI-assisted synthesis without being forced prematurely into programming. For that reason, the programmatic and no-code dimension of this paper is reserved for a single section immediately preceding the conclusion. The earlier sections first establish why AI-assisted evidence synthesis matters methodologically and substantively.

The remainder of the paper proceeds as follows. Section 2 reviews the literature on active learning, LLM-based extraction, open bibliographic APIs, and the methodological status of automated reviews. Section 3 discusses the applicability of these workflows across diverse academic disciplines. Section 4 examines core design principles: query formulation, programmatic literature collection, screening model configuration, and extraction schema design. Section 5 addresses validation, data quality, hallucination risks, and ethical reporting standards (including PRISMA 2020, PRISMA-S, and ROSES). Section 6 discusses technical failure modes specific to AI-assisted workflows. Section 7 considers specific disciplinary applications. Section 8 presents a no-code computational workflow based on repository-hosted Python and R scripts and an interface-based application. Section 9 concludes.

2 Literature Review: AI and Automation in Evidence Synthesis

The methodological literature relevant to AI-assisted systematic reviews spans several adjacent fields rather than a single unified tradition. Understanding these literatures in relation to each other is essential for situating any specific tool or technique within a broader methodological context.

2.1 Machine Learning for Screening

In the domain of machine learning for evidence synthesis, the focus has historically been on the abstract screening phase. Van de Schoot and colleagues introduced ASReview, an open-source platform that implements active learning to accelerate the screening of large textual datasets [Van de Schoot et al., 2021]. Their simulation studies demonstrated that active learning can yield far more efficient reviewing than manual screening while maintaining high recall, often identifying 95% of relevant studies after screening only 5–10% of the corpus. This work established a foundational principle: computational tools should augment rather than replace human judgment. The human reviewer remains in the loop, iteratively labeling a small set of papers to guide a prioritization model that continuously improves its predictions. This principle is central to rigorous review methodology and distinguishes responsible AI-assisted reviewing from fully automated, unsupervised approaches.

Building on this foundation, Teijema and colleagues examined the performance of active learning using deep learning-based models compared to traditional machine learning approaches, finding that switching between models during the review process can improve efficiency [Teijema et al., 2023]. This line of research underscores that no single algorithmic approach is universally optimal; the choice of model should be guided by the characteristics of the corpus and the nature of the research question.

2.2 Large Language Models in Systematic Reviews

A second, rapidly expanding body of work concerns the application of Large Language Models (LLMs) to systematic reviews. The capabilities of models like GPT-4 and Claude have prompted researchers to explore their utility across multiple stages of the review process, from formulating search queries to screening abstracts and extracting data from full texts.

Mahmoudi et al. critically assessed LLM performance in data extraction for systematic reviews, noting both the potential for high accuracy on structured tasks and the persistent risks of nuanced errors and hallucination on complex, contextually sensitive information [Mahmoudi et al., 2025]. Similarly, Kim et al. explored customized GPT-4 models for extracting study data and assessing risk of bias in systematic reviews, highlighting that while

AI can achieve impressive accuracy on well-defined extraction tasks, human oversight remains indispensable for ensuring the integrity of the final dataset [Kim et al., 2025]. Lieberum and colleagues conducted a scoping review of LLM applications in systematic reviews, concluding that while the technology is rapidly maturing, it is not yet ready for unsupervised use and requires robust validation frameworks [Lieberum et al., 2025].

These contributions are vital because they resist the idea that LLMs are infallible. Rather, they treat LLM outputs as probabilistic representations that require validation against the source material. The risk of hallucination—the generation of plausible but factually incorrect information—is particularly acute in the context of data extraction, where an LLM might invent a sample size or misattribute a finding if the prompt is insufficiently constrained.

2.3 Open Bibliographic APIs and the Infrastructure of Retrieval

A third line of scholarship deals with the infrastructure of literature retrieval. The shift toward open science has elevated platforms like OpenAlex, Crossref, Semantic Scholar, and Europe PMC as primary sources for bibliographic metadata. These platforms offer free, programmatic access to millions of scholarly records via well-documented APIs, representing a significant democratization of literature access.

Delgado-Quirós and Ortega analyzed the completeness of metadata across eight free-access scholarly databases, noting the viability of OpenAlex and Crossref as alternatives to proprietary systems like Web of Science or Scopus, while also identifying gaps in abstract coverage for some disciplines [Delgado-Quirós and Ortega, 2024]. Culbert and colleagues further analyzed OpenAlex's reference coverage compared to Web of Science and Scopus, underscoring its utility for programmatic, API-driven corpus construction [Culbert et al., 2025]. This literature is important because it situates AI screening within the broader context of open data access: an AI model can only synthesize the literature it is fed, making the quality and comprehensiveness of the API retrieval process paramount.

2.4 Reporting Standards and Reproducibility

A fourth literature addresses the methodological standards and reporting guidelines governing systematic reviews. The PRISMA 2020 statement provides the definitive framework for reporting systematic reviews, emphasizing transparency in search strategies, study selection, data extraction, and synthesis [Page et al., 2021]. The PRISMA-S extension specifically addresses the rigorous reporting of literature searches, which is critical when using programmatic API retrieval [Rethlefsen et al., 2021]. For environmental and broader life sciences, frameworks such as ROSES (RepOrting standards for Systematic Evidence Syntheses) serve a similar function [Pullin et al., 2018]. As AI tools are integrated into these workflows, adhering to these guidelines becomes critical for ensuring reproducibility and scientific trust. Hardwicke and colleagues' empirical assessment of transparency and reproducibility practices further underscores the need for open code and data sharing, including the sharing of AI prompts and model configurations used in systematic reviews [Hardwicke et al., 2020].

2.5 Ethics and Epistemic Risk

Finally, there is a growing literature on the ethical and epistemic risks of generative AI in research contexts. Issues of algorithmic bias, hallucination, and data privacy are central concerns. As researchers delegate screening and extraction tasks to LLMs, they must navigate the tension between efficiency and the risk of introducing systematic errors into the synthesized evidence base. The broader internet research ethics literature reinforces the importance of contextual decision-making rather than one-size-fits-all prescriptions, a principle that applies equally to the use of AI in evidence synthesis.

3 The Cross-Disciplinary Imperative for AI-Assisted Synthesis

The practical appeal of AI-assisted systematic reviews derives from the universal challenge of information overload across a wide range of academic disciplines. While the methodology of the systematic review originated in the health sciences, the need to rigorously synthesize literature is now ubiquitous.

In the health and life sciences, researchers must aggregate findings from hundreds of randomized controlled trials, epidemiological studies, or genomic analyses to inform clinical guidelines or public health policy. The sheer volume of daily publications makes manual screening a significant barrier to timely evidence synthesis [Wang et al., 2025]. AI tools, particularly active learning for screening and LLMs for extracting specific trial characteristics (e.g., sample size, dosage, primary outcomes, risk of bias assessments), offer a direct solution to this bottleneck [Guo et al., 2024]. The PRISMA framework provides a well-established reporting standard for these reviews, and AI tools must be deployed in a manner consistent with its requirements.

In the social sciences, evidence synthesis often takes the form of scoping reviews, qualitative meta-syntheses, or systematic reviews of policy interventions, educational outcomes, or psychological phenomena. Here, the literature is frequently characterized by diverse methodologies, varying terminologies, and complex theoretical frameworks that resist simple keyword-based retrieval. AI tools can assist in harmonizing these diverse sources, using LLMs to categorize qualitative findings or extract nuanced contextual variables—such as the theoretical framework employed, the country of study, or the specific population targeted—that traditional keyword searches might miss [Fütterer et al., 2026, Garzón et al., 2025].

In the humanities, researchers conduct literature reviews to map the development of theoretical concepts, trace the intellectual genealogy of a scholarly debate, or synthesize interpretive analyses of cultural artifacts. While API-based retrieval and PDF extraction may be less complete for some humanities sub-fields due to the prevalence of monographs and older archival materials, the fundamental need to systematically search, screen, and extract information from a large corpus remains. Open APIs allow for comprehensive mapping of the scholarly literature on a given topic across decades, while LLMs can assist in thematic extraction and narrative synthesis.

In engineering and the applied sciences, researchers conduct systematic reviews to map the state of the art in a specific technology, evaluate the performance of competing methods, or identify gaps in the literature that warrant further investigation. The literature in these fields is often highly technical and quantitative, making structured data extraction a natural application for LLM-based tools, including fully automated end-to-end review systems [Al-Marridi et al., 2025].

What these diverse applications share is not a common substantive domain, but a common workflow of evidence construction. In each case, the researcher begins with a corpus of literature and asks whether AI tools can accurately and efficiently identify relevant studies and extract meaningful data. This commonality is what makes AI-assisted synthesis teachable across a wide range of disciplines.

4 Research Design Principles for AI-Assisted Workflows

4.1 From Research Question to Search Strategy

A robust AI-assisted review begins with a research question rather than with a machine learning model. The literature on systematic reviews consistently warns against source-led or tool-led design. Question-led design determines the search strategy, the selection of bibliographic APIs, and the formulation of Boolean queries. In health sciences, frameworks like PICO (Population, Intervention, Comparison, Outcome) provide a structured approach to translating a clinical question into a search strategy. In the social sciences, analogous frameworks such as SPIDER (Sample, Phenomenon of Interest, Design, Evaluation, Research type) serve a similar function. The choice of framework should reflect the nature of the research question and the type of evidence being sought.

4.2 Programmatic Literature Collection and Corpus Construction

Open APIs such as OpenAlex, Crossref, Semantic Scholar, and Europe PMC offer programmatic access to millions of scholarly records. However, querying these APIs requires careful design. Researchers must translate their conceptual questions into precise search strings, utilizing appropriate filters (e.g., publication year, document type, open access status) and handling pagination to retrieve comprehensive results. The advantage of programmatic literature collection over manual database searching is reproducibility: an API query can be saved, shared, and rerun exactly. However, because APIs evolve and results may change over time, reproducibility requires logging queries and timestamps—a practice that robust workflows must support automatically.

A further advantage of API-based retrieval is the ability to programmatically combine results from multiple sources and apply automated deduplication. Deduplication algorithms that match on DOIs, titles, or a combination of metadata fields can automatically identify and merge these duplicate records, producing a clean, unduplicated corpus ready for screening. The integration of reference management tools, particularly Zotero, adds a further layer of utility, enabling the workflow to push deduplicated records directly into a Zotero collection for review and annotation.

4.3 Configuring the Screening Model

Once a deduplicated corpus is constructed, the researcher must configure the screening approach. This involves a critical choice between Active Learning and LLM-based zero-shot classification.

Active Learning, as implemented in tools like ASReview, requires the researcher to iteratively label a small set of papers to guide a prioritization model. The algorithm trains a model to predict the relevance of the remaining corpus, constantly updating as the researcher provides more labels. This human-in-the-loop approach is highly efficient and maintains strong researcher oversight. The model learns the specific relevance criteria of the current review from the researcher's own judgments. Screening performance is evaluated using recall-oriented metrics to minimize missed relevant studies.

LLM-based screening, conversely, relies on prompting a generative model with explicit inclusion and exclusion criteria. The model evaluates each abstract against these criteria and provides a binary decision, often accompanied by a brief justification. This approach can be faster and requires no initial labeling effort, but it demands rigorous prompt engineering and carries a higher risk of misinterpreting nuanced criteria. The design principle here is that the choice of model must align with the complexity of the inclusion criteria and the acceptable margin of error.

4.4 Extraction Schema Design and Prompt Engineering

The most complex phase of the AI-assisted workflow is data extraction from full-text PDFs. The researcher must first define an explicit, discipline-specific extraction schema—such as a PICO template for health sciences or a Thematic Synthesis template for the social sciences and humanities.

Using LLMs for extraction requires translating this schema into precise, structured prompts. The LLM must be instructed not only on what to extract, but on the required output format (e.g., JSON or a structured table) and the constraints on inference. This is a measurement decision. The prompt defines what counts as an observation and what form of comparability the resulting dataset will support. Recognizing this prompting layer as a site of potential measurement error is essential for responsible interpretation.

5 Validation, Quality, and Ethical Reporting

5.1 The Necessity of Validation

Validation is the cornerstone of AI-assisted evidence synthesis. The fact that an LLM can rapidly populate a spreadsheet does not, by itself, show that the data is accurate or that the screening decisions are correct. AI-assisted extraction is probabilistic and requires auditing. Validation in this context has at least five distinct dimensions.

The first is screening validation: confirming that the active learning model or LLM has not erroneously excluded relevant studies. The second is extraction validation: checking that the LLM has accurately retrieved the requested variables from the full texts without hallucination. This requires the researcher to manually audit a representative subset of the extracted data against the original PDFs. The third is semantic validation: confirming that the extracted values are semantically consistent across studies. The fourth is coverage validation: assessing whether the API retrieval strategy captured a sufficiently comprehensive set of relevant literature. The fifth is analytic validation: determining whether the synthesized data behaves plausibly in summaries, distributions, or meta-analyses.

5.2 Mitigating Hallucination and Algorithmic Bias

Generative AI models are prone to hallucination—generating plausible but incorrect information. In the context of data extraction, an LLM might invent a sample size, misattribute a finding to the wrong study arm, or confabulate a geographic location if the prompt is not sufficiently constrained. Mitigating this risk requires strict prompt engineering, the use of lower temperature settings, and human verification of a representative sample of the extracted data.

Algorithmic bias is another significant concern. If an active learning model is trained on a biased seed set, the model will perpetuate that bias in its screening predictions. Researchers must therefore ensure that their initial training data is deliberately diverse and representative of the full scope of the literature they intend to capture.

5.3 Ethical Reporting and PRISMA 2020

The integration of AI into systematic reviews necessitates transparent reporting. The PRISMA 2020 statement provides a comprehensive checklist for reporting the methods and findings of systematic reviews [Page et al., 2021]. The PRISMA-S extension provides critical guidance for reporting literature searches, which is essential when using API retrieval [Rethlefsen et al., 2021]. For scoping reviews and systematic reviews in broader fields, the ROSES framework provides analogous standards [Pullin et al., 2018]. When AI tools are used, researchers must explicitly report which algorithms or models were employed, how prompts were structured, what training data was used for active learning models, and what validation steps were taken to verify the AI's outputs.

6 Technical Failure Modes in AI-Assisted Workflows

The methodological discussion of AI-assisted systematic reviews often emphasizes research design, validation, and ethical considerations. However, in practice, a substantial share of error arises from technical failure modes that do not necessarily produce explicit warnings but nonetheless affect the integrity of the resulting dataset. Recognizing these failure modes is essential for assessing data quality and for designing robust workflows.

A first category of failure concerns the instability of API endpoints and data schemas. Bibliographic APIs are not static; their data models, query syntax, and rate limits can change over time. A workflow that functions correctly today may produce different results or fail entirely after an API update.

A second category concerns the incomplete coverage of open APIs. No single open API indexes the entire scholarly literature. Certain disciplines, languages, and publication types are systematically underrepresented.

A third category concerns the quality and consistency of LLM outputs. Even with carefully designed prompts, LLM outputs are non-deterministic and may vary across runs, requiring consistency checks. The same paper may yield different extracted values on different runs of the same prompt, introducing a form of non-deterministic measurement error. Researchers should run extraction multiple times on a sample of papers and assess the consistency of the outputs before applying the workflow to the full corpus.

A fourth category concerns PDF parsing failures. AI-assisted extraction may fail due to PDF structure variability, OCR noise, multi-column layouts, or ambiguous phrasing;

therefore all outputs must be treated as provisional and subject to manual verification. When the PDF parser fails to extract clean text, the LLM receives garbled or incomplete input, leading to unreliable extraction outputs or hallucinated extractions based on fragmented sentences.

These technical failure modes reinforce the broader argument that AI-assisted systematic reviewing is not merely a mechanical automation step but a complex measurement process. Incorporating awareness of these risks into research design requires explicit documentation of collection procedures, iterative validation, and human verification at every stage.

7 Disciplinary Applications

A key advantage of AI-assisted evidence synthesis is that it can support substantively diverse projects while relying on a common methodological logic. The workflow described in this paper is not tied to any single discipline; it is a general-purpose research infrastructure that can be adapted to the specific needs of any field.

In the health and life sciences, AI-assisted reviews are most immediately applicable to the synthesis of clinical trial evidence. Researchers can use open APIs to retrieve trial registrations and published results, iteratively label a small set of papers to guide a prioritization model for screening, and use LLMs to extract structured data on interventions, comparators, outcomes, and risk of bias using a PICO template.

In the social sciences, AI-assisted synthesis is particularly valuable for scoping reviews and systematic reviews of policy interventions. Researchers can use programmatic literature collection to retrieve literature on a specific policy domain, apply LLM-based screening to identify studies that evaluate the effectiveness of a particular intervention, and use a Thematic Synthesis template to pull data on study design, population, context, and outcomes.

In the humanities, AI-assisted synthesis can support intellectual history and bibliometric analysis. Researchers can use open APIs to construct comprehensive corpora of publications on a specific theoretical concept or historical period, apply LLMs to extract thematic and argumentative content, and use tools like bibliometrix to map the evolution of scholarly debates over time [Aria and Cuccurullo, 2017].

In engineering and the applied sciences, AI-assisted synthesis is well-suited to systematic reviews of technical performance. Researchers can use open APIs to retrieve literature on a specific technology or method, apply active learning to screen for papers that report quantitative performance metrics, and use structured LLM extraction to pull data on experimental conditions, datasets, and performance results.

8 A No-Code Computational Workflow for Research Training

The previous sections have argued that AI-assisted systematic reviewing is best understood as a rigorous methodological workflow requiring careful design and validation. This section turns to the pedagogical question of how such a workflow can be taught to researchers with little or no programming background.

The central premise is that a no-code seminar should not hide the underlying logic of AI and data collection. Rather, it should separate conceptual learning from the immediate burden of writing code. One practical way to do so is to host the underlying computational procedures in a public repository containing parallel scripts in Python and R, while exposing those procedures to participants through an interface-based application, such as Streamlit. The repository provides inspectable procedures, reproducible versioning, and future extensibility; the application provides accessibility through a guided, no-code interface.

A pedagogically sound application should be modular, reflecting the stages of a systematic review. Instead of forcing all users into a single start point, it should allow entry from different stages of the workflow.

The first module handles **Search and Collect**. Participants input Boolean queries, select target APIs (e.g., OpenAlex, Crossref, Semantic Scholar), and retrieve bibliographic metadata. The interface translates their inputs into API calls, handling pagination and rate-limiting in the background.

The second module addresses **Deduplication and Corpus Management**. The app automatically identifies and merges duplicate records based on DOIs or title-matching algorithms, producing a clean, unduplicated corpus.

The third module supports **AI-Assisted Screening**. Here, users can choose to initiate an Active Learning session—iteratively labeling a training set while the app reorders the list to surface the most likely relevant studies—or configure an LLM zero-shot prompt by typing their inclusion and exclusion criteria into a text box. The app includes a Transparency Log feature that records every prompt used and every AI justification given, providing a full audit trail for reporting purposes.

The fourth module enables **Data Extraction**. Participants upload full-text PDFs and define their extraction schema via the interface, selecting from discipline-specific templates (PICO, Thematic Synthesis) or custom variables. The app sends the extracted text and the schema to the LLM via API, returning a structured data table.

The fifth module facilitates **Synthesis and Map**. Users can generate automated PRISMA flow diagrams based on the screening data, produce summary visualizations of the extracted variables, and generate a Temporal Analysis chart to identify research trends over time. An Interactive Bibliometric Map visualizes co-author networks and recurring keywords from the retrieved metadata.

The most important feature of such an application is configurable transparency. Participants control the API filters, the screening methods, the extraction schemas, and the validation checks. These interface choices translate methodological decisions into accessible controls.

The role of the code repository remains essential, but so do its limits. Hosting both Python and R scripts serves several purposes. It shows that the no-code interface is grounded in inspectable procedures rather than opaque manipulation. It supports reproducibility by preserving the logic of API queries, deduplication algorithms, and LLM prompts. However, a no-code architecture is best viewed as appropriate for introductory and pedagogical contexts, where the aim is to make the logic of AI-assisted synthesis accessible without claiming that the interface alone can eliminate the need for methodological vigilance or technical maintenance.

9 Conclusion

The integration of Artificial Intelligence into systematic reviews and research synthesis represents a profound and ongoing shift in how academic literature is navigated, evaluated, and synthesized. Active learning, LLMs, and open bibliographic APIs provide researchers across a wide range of disciplines with the tools to overcome the bottlenecks of manual screening and extraction that have long constrained the pace and scope of evidence synthesis. Yet, the scientific value of these tools depends entirely on design choices that precede and follow their application.

This paper has argued that AI-assisted synthesis is a multi-stage workflow requiring rigorous query formulation, transparent model configuration, careful schema design, and mandatory human validation. The relevant literatures on machine learning, text-as-data, open bibliographic infrastructure, and research ethics converge on the point that computational power does not reduce the importance of methodological rigor; it amplifies it. The efficiency gains offered by AI tools are real and significant, but they are accompanied by new risks of hallucination, algorithmic bias, technical failure modes, and reproducibility limits that must be actively managed.

For this reason, the pedagogical challenge is not to promise exhaustive technical coverage in a short training format. It is to provide researchers with a realistic and transparent way to understand and utilize these tools within a rigorous methodological framework. A no-code workflow built on open Python and R scripts and implemented through an interface-based application offers a highly effective approach for introductory settings. Such an environment can democratize access to advanced synthesis methods while explicitly acknowledging the risks and ethical responsibilities that accompany AI-assisted research.

Future work in this area should continue to integrate AI capabilities with domain expertise, develop discipline-specific reporting standards for AI-assisted reviews, and build open-source tools that make the full workflow—from programmatic literature collection to validated synthesis—accessible to researchers regardless of their computational background. The scientific community will increasingly rely on AI to manage the information abundance of contemporary research, but the scientific value of this reliance will continue to depend on how carefully and transparently these tools are deployed.

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